PART 3 OF 5 · SOLUTIONS

Stochastic Calculus, Step by Step

Part 3: Brownian motion and Itô.
Solutions to every exercise



Rates with Bip

STOCHASTIC CALCULUS, STEP BY STEP · PART 3 OF 5 · SOLUTIONS

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Part 3: Brownian motion and the Itô integral. Solutions

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Stochastic Calculus, Step by Step. Part 3: Brownian motion and the Itô integral. Solutions.

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The course's numbers: $S_0 = 100$, $K = 105$, $r = 3\%$ (continuously compounded), $\sigma = 20\%$, $T = 1$ year, and $\mu = r$ in Part 3. Made up for teaching; every random number is simulated with a fixed seed.

This volume holds the full solutions to the exercises and self-test questions in Stochastic Calculus, Step by Step, Part 3. Try each exercise before reading its solution.

Every number in the solutions is computed by a script that checks itself with assertions; the code solutions are shown in full.

This book is for education only. It is not financial, investment, legal or tax advice.

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How to use the solutions

This volume holds the full solutions to *Stochastic Calculus, Step by Step, Part 3: Brownian motion and the Itô integral*: every exercise worked out, and the answers to the self-test questions at the end of each lesson. It is meant to be read **after** you have tried each one.

- **By-hand exercises** (the first six of each lesson) are worked in full. Each solution starts with **the idea** (the one thought that unlocks it), then goes through numbered steps with the reason for each, and ends with a check or an interpretation. If you are stuck, read only the idea and try again before reading the steps.
- **Code exercises** (the last two of each lesson) show a complete Python solution and exactly what it prints. Every number in this volume is produced by a script that checks itself with assertions, using fixed random seeds, so your own run should match to the digits shown when you use the same seeds.
- **Self-test answers** are short: a sentence or two, and a number where there is one.

The course's numbers are the same throughout: $S_0 = 100$, $K = 105$, $r = 3\%$ (continuously compounded), $\sigma = 20\%$, $T = 1$ year, and $\mu = r$ in Part 3.

Almost every by-hand solution in Part 3 uses one of three moves: **split** $W_t = W_s + (W_t - W_s)$ and use independence; apply **Itô's lemma** and replace $(dW)^2$ by dt; or **take expectations**, which kills every dW term. If your answer differs, check the usual suspects: a variance written where a standard deviation belongs, a missing $\frac{1}{2}$ in Itô's correction, $(dX)^2 = dt$ written where it should be $b^2\,dt$, and a forgotten f_t term.

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SOLUTIONS TO LESSON 13

Brownian Motion

Solutions to Exercises 13.1 to 13.8 and the Lesson 13 self-test.

Exercise 13.1

The idea. The position is a sum of nt independent steps; means add, and so do variances.

Step 1. One step. δX_i with $\delta = 1/\sqrt{n}$ and $X_i = \pm 1$: mean $\delta\,(\frac{1}{2} - \frac{1}{2}) = 0$, and since $(\delta X_i)^2 = \delta^2 = 1/n$ always, variance $1/n$.

Step 2. Add nt steps. Mean 0; variance $nt \times 1/n = t$ (independent steps: variances add).

Why $1/\sqrt{n}$. With step δ the variance at time t is $nt\delta^2$. It stays at t for every n only if $\delta^2 = 1/n$. With $\delta = 1/n$ the variance would be $t/n \to 0$: the walk would shrink to a flat line at 0, with no randomness left. With $\delta = 1$ it would be $nt \to \infty$.

Exercise 13.2

The idea. Split the later value into the earlier value plus an independent move.

Covariance. For $s \le t$: $W_t = W_s + (W_t - W_s)$, so $E[W_s W_t] = E[W_s^2] + E[W_s]E[W_t - W_s] = s + 0$. Both means are 0, so $\mathrm{Cov}(W_s, W_t) = s = \min(s,t)$.

$\mathrm{Var}(W_1 + W_2).$ $= \mathrm{Var}(W_1) + \mathrm{Var}(W_2) + 2\mathrm{Cov}(W_1, W_2) = 1 + 2 + 2\min(1,2) = 5.$

Correlation of W_1 and W_4. $\dfrac{\min(1,4)}{\sqrt{1}\,\sqrt{4}} = \dfrac{1}{2} = 0.5.$

Exercise 13.3

The idea. The first event is about two independent moves; the second is not.

First probability. W_1 and $W_2 - W_1$ are independent, each normal with mean 0, so each is positive with probability $\frac{1}{2}$:

$$P(W_1 > 0,\ W_2 - W_1 > 0) = \tfrac{1}{2} \times \tfrac{1}{2} = 0.25.$$

Why the second is bigger. $\{W_1 > 0, W_2 > 0\}$ contains every path in the first event (if $W_1 > 0$ and the path then rises, $W_2 > 0$), **plus** the paths that fall over $[1, 2]$ but by less than W_1, and so stay positive. So it is larger: $\frac{3}{8} = 0.375$. The values W_1 and W_2 are positively correlated (correlation $1/\sqrt{2}$), so "both positive" is more likely than for independent variables. (The $\frac{3}{8}$ comes from a standard formula for two correlated normals, $\frac{1}{4} + \arcsin(\rho)/(2\pi)$, with $\rho = 1/\sqrt{2}$, $\arcsin \rho = \pi/4$.)

Exercise 13.4

The idea. Each property of B follows from the same property of W, with times stretched by c.

1. **Start.** $B_0 = W_0/\sqrt{c} = 0$.
2. **Independent increments.** For $t_0 < t_1 < \cdots < t_k$, the increments $B_{t_j} - B_{t_{j-1}} = (W_{ct_j} - W_{ct_{j-1}})/\sqrt{c}$ are increments of W over the non-overlapping intervals $[ct_{j-1}, ct_j]$, hence independent; dividing by a constant keeps them independent.
3. **Normal increments.** $W_{ct} - W_{cs} \sim N(0, c(t-s))$, so dividing by $\sqrt{c}$ gives variance $c(t-s)/c = t - s$: $B_t - B_s \sim N(0, t-s)$.
4. **Continuity.** $t \mapsto W_{ct}/\sqrt{c}$ is a continuous function composed with a stretch: continuous.

So B is a Brownian motion.

Exercise 13.5

The idea. Split, square, and condition term by term.

Step 1. With $D = W_t - W_s$: $W_t^2 = W_s^2 + 2W_sD + D^2$.

Step 2. Given $\mathcal{F}_s$: W_s^2 is known; $E[W_sD \mid \mathcal{F}_s] = W_sE[D] = 0$ (take out what is known, then independence); $E[D^2 \mid \mathcal{F}_s] = E[D^2] = t - s$.

Step 3. $E[W_t^2 \mid \mathcal{F}_s] = W_s^2 + t - s$, so $E[W_t^2 - t \mid \mathcal{F}_s] = W_s^2 - s$. ✓

Exercise 13.6

The idea. Split the exponent; the known part comes out, the independent part has a known average.

Step 1. $Z_t = e^{\sigma W_s - \frac{1}{2}\sigma^2 t}\, e^{\sigma D}$ with $D = W_t - W_s \sim N(0, t-s)$.

Step 2. $E[e^{\sigma D} \mid \mathcal{F}_s] = E[e^{\sigma D}] = e^{\sigma^2(t-s)/2}$ (Lesson 4 with $u = \sigma$, $v = t - s$).

Step 3. $E[Z_t \mid \mathcal{F}_s] = e^{\sigma W_s - \frac{1}{2}\sigma^2 t + \frac{1}{2}\sigma^2(t-s)} = e^{\sigma W_s - \frac{1}{2}\sigma^2 s} = Z_s$. ✓

The share. $e^{-rt}S_t = S_0 e^{(r-\frac{1}{2}\sigma^2)t - rt + \sigma W_t} = S_0 e^{\sigma W_t - \frac{1}{2}\sigma^2 t} = S_0 Z_t$, a constant times a martingale, so a martingale. With $\sigma = 20\%$ the correction is $\frac{1}{2}\sigma^2 = 0.02$ a year.

Exercise 13.7

Scaled random walks at time 1, against the normal.

```python
import math

import numpy as np

N = lambda x: 0.5 * (1 + math.erf(x / math.sqrt(2)))
rng = np.random.default_rng(137)
for n in (10, 100, 10_000):
    heads = rng.binomial(n, 0.5, 200_000)         # how many of the n steps went up
    W1 = (2 * heads - n) / math.sqrt(n)           # the walker at time 1
    print(f"n = {n:6d}: P(W1 <= 1) = {np.mean(W1 <= 1):.4f}   (N(1) = {N(1):.4f}); "
          f"possible values spaced {2 / math.sqrt(n):.3f} apart")
```

What it prints.

```
n =     10: P(W1 <= 1) = 0.8295   (N(1) = 0.8413); possible values spaced 0.632 apart
n =    100: P(W1 <= 1) = 0.8648   (N(1) = 0.8413); possible values spaced 0.200 apart
n =  10000: P(W1 <= 1) = 0.8440   (N(1) = 0.8413); possible values spaced 0.020 apart
```

Why small n is off. The walk at time 1 can only take the values $(2k-n)/\sqrt{n}$, spaced $2/\sqrt{n}$ apart: for $n = 10$, 0.632 apart. Its distribution function is a staircase, and $P(W \le 1)$ depends on where the stairs fall relative to 1. For $n = 100$ one possible value is exactly 1 (55 heads), and it carries probability about 0.05, which is why that row overshoots. As n grows the stairs get finer and the answer approaches $N(1) = 0.8413$.

Exercise 13.8

Covariances and the exponential martingale on simulated paths.

```
rng = np.random.default_rng(138)
paths, steps, T, sigma = 50_000, 400, 2.0, 0.2
dt = T / steps
W = np.concatenate([np.zeros((paths, 1)), np.cumsum(rng.normal(0, math.sqrt(dt), (paths,
steps)), axis=1)], axis=1)
at = lambda t: W[:, int(round(t / dt))]
for s, t in ((0.5, 1.0), (0.5, 2.0), (1.0, 1.5), (1.5, 2.0)):
    print(f"Cov(W_{s}, W_{t}) = {np.mean(at(s) * at(t)):.4f}   min(s, t) = {min(s, t)}")
for t in (0.5, 1.0, 2.0):
    Z = np.exp(sigma * at(t) - 0.5 * sigma**2 * t)
    print(f"E[Z_{t}] = {Z.mean():.4f}")
```

What it prints.

```
Cov(W_0.5, W_1.0) = 0.4968   min(s, t) = 0.5
Cov(W_0.5, W_2.0) = 0.4992   min(s, t) = 0.5
Cov(W_1.0, W_1.5) = 0.9898   min(s, t) = 1.0
Cov(W_1.5, W_2.0) = 1.4811   min(s, t) = 1.5
E[Z_0.5] = 1.0012
E[Z_1.0] = 1.0014
E[Z_2.0] = 1.0028
```

Reading it. Every covariance is within about 0.02 of $\min(s, t)$, the size of sampling noise with 50,000 paths. $E[Z_t]$ stays at 1 for all three times: the exponential martingale has constant expectation.

Self-test

1. Variance 0.25, standard deviation 0.5.

2. So that the variance at time t is exactly t for every n: smaller steps would make the walk vanish, bigger ones would make it explode.

3. Only W_t. Fixed versions: $W_t^2 - t$ and $e^{W_t - t/2}$.

SOLUTIONS TO LESSON 14

The Strange Properties of Brownian Paths

Solutions to Exercises 14.1 to 14.8 and the Lesson 14 self-test.

Exercise 14.1

The idea. Dividing a normal by a constant scales its standard deviation.

Step 1. $W_{t+h} - W_t \sim N(0, h)$, so $Q_h = (W_{t+h} - W_t)/h \sim N(0, h/h^2) = N(0, 1/h)$.

Step 2. $Q_h = Z/\sqrt{h}$ with Z standard normal, so $P(|Q_h| > 1) = P(|Z| > \sqrt{h}) = 2\big(1 - N(\sqrt{h})\big)$.

- $h = 0.01$: $2(1 - N(0.1)) = 0.9203$.
- $h = 0.0001$: $2(1 - N(0.01)) = 0.9920$.

Interpretation. As the step shrinks, the slope is ever more likely to exceed 1 (or any bound): it does not settle.

Exercise 14.2

The idea. Each move is $\sqrt{\Delta t}$ times a standard normal, and $E|Z| = \sqrt{2/\pi}$.

$$E\sum_{i=1}^{n} |\Delta W_i| = n\sqrt{\Delta t}\sqrt{\frac{2}{\pi}} = n\sqrt{\frac{T}{n}}\sqrt{\frac{2}{\pi}} = \sqrt{\frac{2nT}{\pi}}.$$

With $T = 1$: 7.98 for $n = 100$, 79.79 for $n = 10{,}000$, 797.9 for $n = 1{,}000{,}000$. Each hundredfold increase in n makes the path ten times longer.

Exercise 14.3

The idea. Mean to T, variance to 0, then Chebyshev.

Mean. $E[(\Delta W_i)^2] = \Delta t$, so $E[Q_n] = n\Delta t = T$.

Variance. $(\Delta W_i)^2 = \Delta t\, Z_i^2$, and $\mathrm{Var}(Z^2) = E[Z^4] - (E[Z^2])^2 = 3 - 1 = 2$. So each square has variance $2\Delta t^2$; they are independent, so $\mathrm{Var}(Q_n) = 2n\Delta t^2 = 2T^2/n$.

Convergence. By Chebyshev (Lesson 6), for any $\varepsilon > 0$:

$$P(|Q_n - T| \geq \varepsilon) \leq \frac{2T^2}{n\varepsilon^2} \to 0.$$

Exercise 14.4

The idea. Count orders of magnitude: each term is tiny, and there are not enough of them.

$\sum \Delta W_i\, \Delta t$. Mean $\Delta t \sum E[\Delta W_i] = 0$. Variance $\Delta t^2 \sum \mathrm{Var}(\Delta W_i) = \Delta t^2 \cdot n\Delta t = T\Delta t^2 \to 0$. Mean 0 and variance to 0: it tends to 0.

$\sum \Delta t^2 = n\Delta t^2 = T\Delta t \to 0$.

Why $\sum(\Delta W_i)^2$ is different. Each term is of order Δt (not $\Delta t^{3/2}$ or Δt^2), and there are $T/\Delta t$ terms: the total is of order 1. In the other two sums each term is of a higher order than Δt, so $T/\Delta t$ of them still add up to something that vanishes.

Exercise 14.5

The idea. Smooth parts contribute nothing; each unit of dW contributes dt, scaled by the square of its coefficient.

(a) $f(t) = 100e^{0.03t}$ is smooth (its slope is at most $3e^{0.03T}$), so its quadratic variation is 0.

(b) $\sum(\sigma\Delta W_i)^2 = \sigma^2 \sum(\Delta W_i)^2 \to \sigma^2 T$.

(c) $\Delta \ln S_i = (r - \frac{1}{2}\sigma^2)\Delta t + \sigma\Delta W_i$. Squaring: $(r - \frac{1}{2}\sigma^2)^2\Delta t^2 + 2(r - \frac{1}{2}\sigma^2)\sigma\Delta t\, \Delta W_i + \sigma^2(\Delta W_i)^2$. By Exercise 14.4 the first two sums vanish; the third tends to $\sigma^2 T$. So $[\ln S]_T = \sigma^2 T$: with $\sigma = 20\%$ and $T = 1$, 0.04. The drift doesn't matter.

Exercise 14.6

The idea. B is a sum of independent normal pieces; then expand the product of moves.

B is a Brownian motion. $B_0 = 0$. Its increment over $[s, t]$ is $\rho(W_t - W_s) + \sqrt{1-\rho^2}(W'_t - W'_s)$: independent normals with variances $\rho^2(t-s)$ and $(1-\rho^2)(t-s)$, so normal with variance $t - s$. Increments over separate intervals involve separate pieces of W and W', so they are independent. And B is continuous.

The cross sum. $\Delta W_i\,\Delta B_i = \rho(\Delta W_i)^2 + \sqrt{1-\rho^2}\,\Delta W_i\,\Delta W_i'$. The first part sums to ρT (Lesson 14). The second has mean 0 (ΔW_i and $\Delta W_i'$ independent, both mean 0) and variance $(1-\rho^2)\sum \Delta t^2 = (1-\rho^2)T\Delta t \to 0$. So $\sum \Delta W_i\,\Delta B_i \to \rho T$: $dW\,dB = \rho\,dt$.

Exercise 14.7

Realised variance of one year of daily log returns, 10,000 times.

```
import numpy as np

S0, r, sigma, T, n = 100.0, 0.03, 0.20, 1.0, 252
dt = T / n
rng = np.random.default_rng(147)
dW = rng.normal(0, math.sqrt(dt), (10_000, n))
dlogS = (r - 0.5 * sigma**2) * dt + sigma * dW          # daily log returns
rv = (dlogS**2).sum(axis=1)                             # realised variance, one per year
print(f"realised variance: mean {rv.mean():.5f}, sd {rv.std():.5f}")
print(f"sigma^2 T = {sigma**2 * T:.5f}; theory sd sigma^2 sqrt(2T dt) = {sigma**2 *
math.sqrt(2 * T * dt):.5f}")
print(f"realised volatility sqrt(rv): mean {np.sqrt(rv).mean():.4f}")
```

What it prints.

```
realised variance: mean 0.03999, sd 0.00358
sigma^2 T = 0.04000; theory sd sigma^2 sqrt(2T dt) = 0.00356
realised volatility sqrt(rv): mean 0.1998
```

Reading it. The realised variance averages $\sigma^2 T = 0.04$, with a standard deviation of 0.00358, matching $\sigma^2\sqrt{2T\Delta t}$ (Exercise 14.3 with variance σ^2 per unit time). So one year of daily prices measures the variance to within about 9%, and the volatility to within about 4.5%. With intraday data the error shrinks further: in theory, to 0.

Exercise 14.8

First and quadratic variation of one path, refined.

```
rng = np.random.default_rng(148)
fine = 2**20
W = np.concatenate([[0], np.cumsum(rng.normal(0, math.sqrt(1 / fine), fine))])
print("   steps   first variation   quadratic variation")
for k in range(4, 21, 2):
    incr = np.diff(W[:: fine // 2**k])                  # the same path, cut into 2^k steps
    print(f"{2**k:8d}   {np.abs(incr).sum():15.2f}   {(incr**2).sum():19.4f}")
```

What it prints.

```
   steps   first variation   quadratic variation
      16              3.89                1.6096
      64              6.30                1.1041
     256             12.73                0.9775
    1024             25.40                0.9939
    4096             51.04                1.0142
   16384            101.95                0.9914
   65536            203.30                0.9934
  262144            408.54                0.9983
 1048576            816.64                0.9992
```

Reading it. Each time the number of steps goes up four times, the first variation doubles (it grows like $\sqrt{n}$), while the quadratic variation settles near 1 and its wobble shrinks. Plotted on a logarithmic axis, the first is a straight line with slope ½ and the second a flat line at T.

Self-test

1. The move over a step h has size about $\sqrt{h}$, so the slope has size about $\sqrt{h}/h = 1/\sqrt{h}$, which grows without limit as $h \to 0$.

2. $\sigma^2 T = 0.04$.

3. 1. The quadratic variation is $T = 1$ on every path, whatever W_1 turned out to be.

SOLUTIONS TO LESSON 15

The Itô Integral

Solutions to Exercises 15.1 to 15.8 and the Lesson 15 self-test.

Exercise 15.1

The idea. Expand the square of the next value; the sum of the differences telescopes.

Step 1. For each step, $W_{t_{i+1}}^2 = (W_{t_i} + \Delta W_i)^2 = W_{t_i}^2 + 2W_{t_i}\Delta W_i + (\Delta W_i)^2$, so

$$W_{t_i}\Delta W_i = \tfrac{1}{2}\big(W_{t_{i+1}}^2 - W_{t_i}^2\big) - \tfrac{1}{2}(\Delta W_i)^2.$$

Step 2. Add over i. The first bracket telescopes: $\sum_i (W_{t_{i+1}}^2 - W_{t_i}^2) = W_T^2 - W_0^2 = W_T^2$. So

$$\sum_i W_{t_i}\Delta W_i = \tfrac{1}{2}W_T^2 - \tfrac{1}{2}\sum_i (\Delta W_i)^2,$$

exactly, for any partition.

Step 3. Limit. $\sum(\Delta W_i)^2 \to T$ (Lesson 14), so the left sums tend to $\tfrac{1}{2}W_T^2 - \tfrac{1}{2}T$.

Exercise 15.2

The idea. Right sum = left sum + quadratic variation.

$$\sum W_{t_{i+1}}\Delta W_i = \sum W_{t_i}\Delta W_i + \sum(\Delta W_i)^2 \to \big(\tfrac{1}{2}W_T^2 - \tfrac{1}{2}T\big) + T = \tfrac{1}{2}W_T^2 + \tfrac{1}{2}T.$$

Why no trader earns it. It holds $W_{t_{i+1}}$ shares during $[t_i, t_{i+1}]$: the number of shares depends on the price at the **end** of the step, which is not known when the position must be taken. A trader who could do that would know each move in advance, and would make $E[\cdot] = T > 0$ from a fair game: an arbitrage built on clairvoyance.

Exercise 15.3

The idea. Mean from the martingale property, variance from the isometry, then check directly.

Mean. $E\int_0^T W\,dW = 0$ (martingale, W adapted).

Variance. By the isometry, $E\big[(\int_0^T W\,dW)^2\big] = E\int_0^T W_t^2\,dt = \int_0^T E[W_t^2]\,dt = \int_0^T t\,dt = \frac{1}{2}T^2$. (Swapping E and $\int dt$ is allowed: both are sums.) The mean is 0, so this is the variance.

Direct check. $\int_0^T W\,dW = \frac{1}{2}(W_T^2 - T)$:

- mean $\frac{1}{2}(E[W_T^2] - T) = \frac{1}{2}(T - T) = 0$ ✓;
- variance $\frac{1}{4}\mathrm{Var}(W_T^2) = \frac{1}{4}(E[W_T^4] - (E[W_T^2])^2) = \frac{1}{4}(3T^2 - T^2) = \frac{1}{2}T^2$ ✓.

Exercise 15.4

The idea. A deterministic integrand gives a sum of independent normals.

Step 1. $\sum t_i \Delta W_i$ is a sum of independent $N(0, t_i^2\Delta t)$ variables, so it is normal with mean 0 and variance $\sum t_i^2\Delta t \to \int_0^1 t^2\,dt = \frac{1}{3}$. Limits of normals with converging variances are normal: $J \sim N(0, \frac{1}{3})$.

Step 2. $P(J > 0.5) = P\big(Z > 0.5/\sqrt{1/3}\big) = 1 - N(0.8660) = 0.1932$.

Exercise 15.5

The idea. Both are integrals of ordinary functions against the same dW.

$W_1 = \int_0^1 1\,dW_t$ and $J = \int_0^1 t\,dW_t$, both with mean 0. So

$$\mathrm{Cov}(W_1, J) = E\Big[\int_0^1 1\,dW \int_0^1 t\,dW\Big] = \int_0^1 1\cdot t\,dt = \tfrac{1}{2}.$$

Why the product formula holds. Apply the isometry to $f+g$ and to $f-g$, and subtract: $E[(\int(f+g)dW)^2] - E[(\int(f-g)dW)^2] = 4E[\int f\,dW \int g\,dW]$, while the right sides give $\int(f+g)^2 - \int(f-g)^2 = 4\int fg$.

Check in the discrete sums. $E\big[\sum_i \Delta W_i \sum_j t_j \Delta W_j\big] = \sum_i t_i E[(\Delta W_i)^2] = \sum t_i \Delta t \to \frac{1}{2}$ (cross terms $i \neq j$ have mean 0).

Exercise 15.6

The idea. Sum the discrete product rule over the steps; it telescopes.

Step 1. Check the identity. $t_{i+1}W_{t_{i+1}} - t_i W_{t_i} = t_i(W_{t_{i+1}} - W_{t_i}) + (t_{i+1} - t_i)W_{t_{i+1}} = t_i\Delta W_i + W_{t_{i+1}}\Delta t$. (Add and subtract $t_i W_{t_{i+1}}$.)

Step 2. Add over i. The left side telescopes to $TW_T - 0$:

$$TW_T = \sum_i t_i \Delta W_i + \sum_i W_{t_{i+1}} \Delta t.$$

Step 3. Limits. The first sum tends to $\int_0^T t\, dW_t$ (left endpoints, as required). The second is a Riemann sum for the ordinary integral $\int_0^T W_t\, dt$ (for a continuous path the endpoint doesn't matter here: there is no dW in it). So $\int_0^T t\, dW_t = TW_T - \int_0^T W_t\, dt$.

Note. No extra term appears: t is smooth, so the product's "cross term" $\Delta t\, \Delta W$ sums to 0. Lesson 18's product rule says the same.

Exercise 15.7

Left, right and average sums for ∫W dW on one path.

```
import numpy as np

rng = np.random.default_rng(157)
fine = 2**18
W = np.concatenate([[0], np.cumsum(rng.normal(0, math.sqrt(1 / fine), fine))])
W1 = W[-1]
print(f"W_1 = {W1:.4f}: (W^2 - 1)/2 = {(W1**2 - 1) / 2:.4f}, (W^2 + 1)/2 = {(W1**2 + 1) /
2:.4f}, W^2/2 = {W1**2 / 2:.4f}")
print("   steps      left     right    middle")
for k in range(4, 19, 2):
    Wk = W[:: fine // 2**k]                    # the same path on 2^k steps
    d = np.diff(Wk)
    left, right = (Wk[:-1] * d).sum(), (Wk[1:] * d).sum()
    print(f"{2**k:8d}  {left:8.4f}  {right:8.4f}  {(left + right) / 2:8.4f}")
```

What it prints.

```
W_1 = 1.0243: (W^2 - 1)/2 = 0.0246, (W^2 + 1)/2 = 1.0246, W^2/2 = 0.5246
   steps      left     right    middle
      16    0.1732    0.8760    0.5246
      64   -0.0039    1.0531    0.5246
     256    0.0768    0.9724    0.5246
    1024    0.0470    1.0022    0.5246
    4096    0.0370    1.0122    0.5246
   16384    0.0212    1.0280    0.5246
   65536    0.0267    1.0225    0.5246
  262144    0.0252    1.0241    0.5246
```

Reading it. On this path $W_1 = 1.0243$. The average column is $\frac{1}{2}W_1^2$ exactly, at every n (it telescopes). The left and right columns wander for small n and close in on $\frac{1}{2}(W_1^2 - 1)$ and $\frac{1}{2}(W_1^2 + 1)$ as the steps shrink, always 1 apart in the limit.

Exercise 15.8

100,000 samples of two Itô integrals.

```
paths, steps = 100_000, 500
dt = 1 / steps
t = np.arange(steps) * dt                       # left endpoints
J = np.zeros(paths); I = np.zeros(paths); W = np.zeros(paths)
for k in range(steps):                          # step through time to keep memory small
    dW = rng.normal(0, math.sqrt(dt), paths)
    J += t[k] * dW
    I += W * dW                                 # W is still the left-endpoint value here
    W += dW
print(f"int t dW: mean {J.mean():+.4f}, variance {J.var():.4f} (theory 1/3 = {1 / 3:.4f}),
P(< 0) = {np.mean(J < 0):.4f}")
print(f"int W dW: mean {I.mean():+.4f}, variance {I.var():.4f} (theory 1/2), P(< 0) =
{np.mean(I < 0):.4f} "
      f"(theory P(W_1^2 < 1) = {2 * Ncdf(1) - 1:.4f})")
```

What it prints.

```
int t dW: mean +0.0001, variance 0.3344 (theory 1/3 = 0.3333), P(< 0) = 0.4996
int W dW: mean +0.0020, variance 0.5048 (theory 1/2), P(< 0) = 0.6812 (theory P(W_1^2 < 1) =
0.6827)
```

Reading it. $\int t\,dW$ is symmetric (normal), so it is negative half the time; its variance is about $\frac{1}{3}$. $\int W\,dW = \frac{1}{2}(W_1^2 - 1)$ is lopsided: negative whenever $|W_1| < 1$, which has probability 0.6827, but with a long right tail that brings the mean back to 0. Same mean, very different shapes.

Self-test

1. Because the integrand is a trading position, chosen before the move it earns on. Left endpoints keep it adapted, and that is what makes the integral a martingale.

2. Mean 0 and variance $\frac{1}{2}$.

3. $\int_0^2 3\,dW_t = 3W_2 \sim N(0, 9 \times 2) = N(0, 18)$.

SOLUTIONS TO LESSON 16

Itô's Lemma

Solutions to Exercises 16.1 to 16.8 and the Lesson 16 self-test.

Exercise 16.1

The idea. Apply $df = f'\,dW + \frac{1}{2}f''\,dt$; in expectations, the dW part vanishes.

W^3. $f' = 3x^2$, $f'' = 6x$: $d(W^3) = 3W^2\,dW + 3W\,dt$. *Check:* $E[W_T^3] - 0 = 0 + \int_0^T 3E[W_t]\,dt = 0$, and indeed $E[W_T^3] = 0$ (an odd moment of a symmetric distribution).

e^W. $f' = f'' = e^x$: $d(e^W) = e^W\,dW + \frac{1}{2}e^W\,dt$. *Check:* $E[e^{W_T}] - 1 = \int_0^T \frac{1}{2}E[e^{W_t}]\,dt$. With $E[e^{W_t}] = e^{t/2}$ (Lesson 4), the right side is $\int_0^T \frac{1}{2}e^{t/2}\,dt = e^{T/2} - 1$. ✓

Exercise 16.2

The idea. Use the two-variable form with $f(t, x) = tx$ and input $X = W$ ($a = 0, b = 1$).

$f_t = x$, $f_x = t$, $f_{xx} = 0$, so

$$d(tW_t) = (W_t + 0 + \tfrac{1}{2} \cdot 1 \cdot 0)\,dt + t\,dW_t = W_t\,dt + t\,dW_t.$$

Ordinary calculus (the product rule) gives the same, $d(tW) = W\,dt + t\,dW$: **there is no Itô correction**, because f is a straight line in x ($f_{xx} = 0$). The correction needs curvature.

Exercise 16.3

The idea. Two-variable Itô with $f(t, w) = e^{\sigma w - \frac{1}{2}\sigma^2 t}$.

$f_t = -\frac{1}{2}\sigma^2 f$, $f_w = \sigma f$, $f_{ww} = \sigma^2 f$. With input W:

$$dZ = \left(-\tfrac{1}{2}\sigma^2 Z + \tfrac{1}{2}\sigma^2 Z\right)dt + \sigma Z\,dW = \sigma Z\,dW.$$

The time derivative and the Itô correction cancel exactly: no dt term.

Exercise 16.4

The idea. Itô on W^6 links the sixth moment to the fourth.

Step 1. $f = x^6$: $f' = 6x^5$, $f'' = 30x^4$, so $d(W^6) = 6W^5\,dW + 15W^4\,dt$.

Step 2. Take expectations: $E[W_T^6] = \int_0^T 15E[W_t^4]\,dt = \int_0^T 15 \cdot 3t^2\,dt = 15T^3$.

(Lesson 4's $E[Z^6] = 15$ is the case $T = 1$.)

Exercise 16.5

The idea. Itô on $f(x) = x^2$ with the general input; then average and integrate.

Step 1. $f_x = 2x$, $f_{xx} = 2$:

$$d(X^2) = (2aX + \tfrac{1}{2}b^2 \cdot 2)\,dt + 2bX\,dW = (2aX + b^2)\,dt + 2bX\,dW.$$

Step 2. Average: $\frac{d}{dt}E[X_t^2] = 2aE[X_t] + b^2 = 2a \cdot at + b^2$ (since $E[X_t] = at$).

Step 3. Integrate from 0 with $E[X_0^2] = 0$: $E[X_T^2] = a^2T^2 + b^2T$.

Check. $X_T = aT + bW_T$ has mean aT and variance b^2T, so $E[X_T^2] = (aT)^2 + b^2T$. ✓
With the lesson's $a = 0.03$, $b = 0.2$, $T = 1$: 0.0409.

Exercise 16.6

The idea. The cosine's second derivative is minus itself, so its average solves $m' = -\frac{1}{2}m$.

Step 1. $f = \cos x$: $f' = -\sin x$, $f'' = -\cos x$. So $dY = -\sin W\,dW - \frac{1}{2}\cos W\,dt$.

Step 2. Average: $m(t) = E[\cos W_t]$ satisfies $m'(t) = -\frac{1}{2}m(t)$, with $m(0) = \cos 0 = 1$.

Step 3. Solve (Lesson 2): $m(t) = e^{-t/2}$. At $t = 1$: 0.6065.

Interpretation. On average the cosine of a spreading random point decays: the further it spreads, the more its positive and negative values cancel.

Exercise 16.7

Itô's lemma for x^3 on one path.

```
import numpy as np

rng = np.random.default_rng(167)
fine = 2**18
W = np.concatenate([[0], np.cumsum(rng.normal(0, math.sqrt(1 / fine), fine))])
print(f"W_1^3 = {W[-1] ** 3:.5f}")
print("   steps    sum 3W^2 dW    + sum 3W dt    total")
for k in range(6, 19, 2):
    n = 2**k
    Wn = W[:: fine // n]
    first = (3 * Wn[:-1] ** 2 * np.diff(Wn)).sum()
    ito = (3 * Wn[:-1] * (1 / n)).sum()
    print(f"{n:8d}   {first:12.5f}   {ito:12.5f}   {first + ito:8.5f}")
```

What it prints.

```
W_1^3 = 13.21623
   steps    sum 3W^2 dW    + sum 3W dt    total
      64        9.93773        3.81400   13.75173
     256        9.55822        3.87489   13.43311
    1024        9.35323        3.88797   13.24120
    4096        9.45031        3.88991   13.34021
   16384        9.29872        3.89070   13.18942
   65536        9.32862        3.89080   13.21942
  262144        9.33020        3.89085   13.22105
```

Reading it. This path ends high ($W_1^3 = 13.2162$). The Itô term $\sum 3W\Delta t$ is large and settles quickly (it is an ordinary integral); the stochastic sum $\sum 3W^2\Delta W$ wanders more before settling. Their total closes in on W_1^3; without the Itô term it would miss by about $\int_0^1 3W_t\,dt$, which on this path is close to 4.

Exercise 16.8

Two expectations derived with Itô's lemma, by simulation.

```
Z = rng.standard_normal(1_000_000)
for t in (0.5, 1.0, 2.0):
    Wt = math.sqrt(t) * Z                       # W_t has the distribution of sqrt(t) Z
    print(f"t = {t}: E[cos W_t] = {np.cos(Wt).mean():.4f} (e^(-t/2) = {math.exp(-t /
2):.4f}), "
          f"E[W_t^4] = {(Wt**4).mean():.4f} (3t^2 = {3 * t * t:.4f})")
```

What it prints.

```
t = 0.5: E[cos W_t] = 0.7788 (e^(-t/2) = 0.7788), E[W_t^4] = 0.7492 (3t^2 = 0.7500)
t = 1.0: E[cos W_t] = 0.6065 (e^(-t/2) = 0.6065), E[W_t^4] = 2.9966 (3t^2 = 3.0000)
t = 2.0: E[cos W_t] = 0.3677 (e^(-t/2) = 0.3679), E[W_t^4] = 11.9864 (3t^2 = 12.0000)
```

Reading it. Both formulas, derived without any density, match the simulation to three or four decimals.

Self-test

1. $d(W^2) = 2W\,dW + dt$. The dt is $\frac{1}{2}f''(dW)^2 = \frac{1}{2}\cdot 2\cdot dt$: the second-order Taylor term, which survives because $(dW)^2 = dt$.

2. By $\frac{1}{2}\sigma^2 = 0.02$, that is, 2% a year.

3. $b^2\,dt$.

SOLUTIONS TO LESSON 17

Geometric Brownian Motion

Solutions to Exercises 17.1 to 17.8 and the Lesson 17 self-test.

Exercise 17.1

The idea. Itô's lemma on $\ln S$; the S's cancel and leave constants.

Step 1. $f(S) = \ln S$: $f' = 1/S$, $f'' = -1/S^2$, $f_t = 0$.

Step 2. Input $dS = \mu S\,dt + \sigma S\,dW$, so $(dS)^2 = \sigma^2 S^2\,dt$. Then

$$d\ln S = \frac{1}{S}\,dS - \frac{1}{2S^2}(dS)^2 = \mu\,dt + \sigma\,dW - \tfrac{1}{2}\sigma^2\,dt = \left(\mu - \tfrac{1}{2}\sigma^2\right)dt + \sigma\,dW.$$

Step 3. Integrate: $\ln S_t = \ln S_0 + (\mu - \frac{1}{2}\sigma^2)t + \sigma W_t$.

Step 4. Exponentiate: $S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$.

Exercise 17.2

The idea. Differentiate the candidate with the two-variable Itô's lemma and compare.

With $f(t, w) = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma w}$: $f_t = (\mu - \frac{1}{2}\sigma^2)f$, $f_w = \sigma f$, $f_{ww} = \sigma^2 f$. Input W ($a = 0$, $b = 1$):

$$dS = \left((\mu - \tfrac{1}{2}\sigma^2)S + \tfrac{1}{2}\sigma^2 S\right)dt + \sigma S\,dW = \mu S\,dt + \sigma S\,dW.$$

✓ And S_0 is right at $t = 0$.

Exercise 17.3

The idea. $\ln S_T \sim N(m, s^2)$ with $m = \ln S_0 + (\mu - \frac{1}{2}\sigma^2)T$, $s = \sigma\sqrt{T}$; use Lesson 5.

- **Mean** $S_0 e^{\mu T} = 103.0455$.
- **Median** $e^m = S_0 e^{(\mu - \frac{1}{2}\sigma^2)T} = 101.0050$.
- **Mode** $e^{m - s^2} = S_0 e^{(\mu - \frac{3}{2}\sigma^2)T} = 97.0446$.

$P(S_T > E[S_T])$. $S_T > S_0 e^{\mu T} \iff (\mu - \frac{1}{2}\sigma^2)T + \sigma W_T > \mu T \iff W_T > \frac{1}{2}\sigma T \iff Z > \frac{1}{2}\sigma\sqrt{T}$. So the probability is $N(-\frac{1}{2}\sigma\sqrt{T}) = N(-0.1) = 0.4602$.

Why below a half. The lognormal is skewed to the right: rare very high outcomes pull the mean above the median. The mean is above the middle of the distribution, so less than half the probability lies above it.

Exercise 17.4

The idea. Itô on S^2 gives a differential equation for its mean.

Step 1. $d(S^2) = 2S\,dS + (dS)^2 = (2\mu + \sigma^2)S^2\,dt + 2\sigma S^2\,dW$.

Step 2. $m(t) = E[S_t^2]$: $m' = (2\mu + \sigma^2)m$, $m(0) = S_0^2$, so $m(T) = S_0^2 e^{(2\mu+\sigma^2)T} = 11051.71$ (rate 0.10).

Step 3. $\mathrm{Var}(S_T) = S_0^2 e^{(2\mu+\sigma^2)T} - S_0^2 e^{2\mu T} = S_0^2 e^{2\mu T}(e^{\sigma^2 T} - 1) = 433.34$, standard deviation 20.82. This is Lesson 5's formula (with r there in place of μ).

Exercise 17.5

The idea. It is S_0 times the exponential martingale.

$e^{-\mu t}S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t - \mu t + \sigma W_t} = S_0 e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$, a martingale (Lesson 13).

When $\mu = r$: $e^{-rt}S_t$ is a martingale: the discounted share is a fair game, the defining property of a risk-neutral world (Lessons 11 and 13).

Exercise 17.6

The idea. The log return is normal; standardise.

$\ln(S_T/S_0) \sim N\big((\mu - \frac{1}{2}\sigma^2)T,\ \sigma^2 T\big)$: here mean 0.01 and standard deviation 0.2.

- $P(S_1 < 100) = P(\ln(S_1/S_0) < 0) = N\big((0 - 0.01)/0.2\big) = N(-0.05) = 0.4801$.
- $P(S_1 < 80) = N\big(\ln 0.8 - 0.01)/0.2\big) = N\big((-0.2231 - 0.01)/0.2\big) = N(-1.1657) = 0.1219$.

Exercise 17.7

Strong and weak errors of Euler's method, same noise as the exact solution.

```
import numpy as np

rng = np.random.default_rng(177)
paths, fine = 20_000, 1024
dW = rng.normal(0, math.sqrt(T / fine), (paths, fine))
exact = S0 * np.exp((mu - 0.5 * sigma**2) * T + sigma * dW.sum(axis=1))
print("   steps   strong error   weak error")
for n in (4, 16, 64, 256, 1024):
    dWn = dW.reshape(paths, n, fine // n).sum(axis=2)             # the same noise, n coarser
steps
    euler = S0 * np.prod(1 + mu * T / n + sigma * dWn, axis=1)
    strong = np.abs(euler - exact).mean()
    weak = abs(euler.mean() - exact.mean())                       # same paths, so the sam-
pling noise cancels
    print(f"{n:8d}   {strong:12.4f}   {weak:10.4f}")
```

What it prints.

```
   steps   strong error   weak error
       4         1.1223       0.0176
      16         0.5728       0.0075
      64         0.2904       0.0029
     256         0.1441       0.0023
    1024         0.0719       0.0002
```

Reading it. The strong error halves each time n is multiplied by 4: it shrinks like $n^{-1/2}$. The weak error shrinks faster, roughly like n^{-1}, but on 20,000 paths it is so small that sampling noise shows in the later rows. (The lesson computed the weak error exactly instead, from $E[S_T^{\text{Euler}}] = S_0(1 + \mu T/n)^n$, which avoids the noise.)

Exercise 17.8

Milstein against Euler.

```
print("   steps   Euler strong   Milstein strong")
for n in (4, 16, 64, 256, 1024):
    dWn = dW.reshape(paths, n, fine // n).sum(axis=2)
    h = T / n
    euler = S0 * np.prod(1 + mu * h + sigma * dWn, axis=1)
    milstein = S0 * np.prod(1 + mu * h + sigma * dWn + 0.5 * sigma**2 * (dWn**2 - h), axis=1)
    print(f"{n:8d}   {np.abs(euler - exact).mean():12.4f}   {np.abs(milstein -
exact).mean():15.5f}")
```

What it prints.

```
   steps   Euler strong   Milstein strong
       4         1.1223           0.12029
      16         0.5728           0.03362
      64         0.2904           0.00873
     256         0.1441           0.00220
    1024         0.0719           0.00055
```

Reading it. Milstein's strong error falls by about 4 each time n is multiplied by 4 (the ratio between 256 and 1024 steps is 3.99): strong order 1, against Euler's ½. The extra term $\frac{1}{2}\sigma^2(\Delta W^2 - \Delta t)$ is exactly the piece of the exact growth factor that Euler drops (Lesson 17's last section). With 1024 steps, Milstein is more than 100 times more accurate path by path.

Self-test

1. $\mu - \frac{1}{2}\sigma^2 = 0.01$: 1% a year.

2. The lognormal is skewed right: a few very high outcomes pull the mean up, above the median.

3. The strong error shrinks by about $\sqrt{2}$ (to about 71%); the weak error by about 2 (to about 50%).

SOLUTIONS TO LESSON 18

The Itô Toolbox

Solutions to Exercises 18.1 to 18.8 and the Lesson 18 self-test.

Exercise 18.1

The idea. Itô's lemma, then look only at the dt term.

PROCESS	ITÔ'S LEMMA	DT TERM	MARTINGALE?
W^3	$3W^2\,dW + 3W\,dt$	$3W$	no
$W^3 - 3tW$	$3(W^2 - t)\,dW$	0	yes
e^W	$e^W\,dW + \frac{1}{2}e^W\,dt$	$\frac{1}{2}e^W$	no
$e^{W-t/2}$	$e^{W-t/2}\,dW$	0	yes
tW	$W\,dt + t\,dW$	W	no

For $W^3 - 3tW$: the term $-3tW$ has $d(tW) = W\,dt + t\,dW$ (product rule), so its $-3W\,dt$ cancels the $3W\,dt$ from W^3. For $e^{W-t/2}$: the time derivative gives $-\frac{1}{2}e^{W-t/2}\,dt$, which cancels the Itô term $+\frac{1}{2}e^{W-t/2}\,dt$.

Exercise 18.2

The idea. The exponent's $-ct$ must cancel the curvature term $\frac{1}{2}\theta^2$.

With $f(t, w) = e^{\theta w - ct}$: $f_t = -cf$, $f_{ww} = \theta^2 f$, so the dt term is $(-c + \frac{1}{2}\theta^2)f$. It vanishes when $c = \frac{1}{2}\theta^2$. For $\theta = 2$: $c = 2$.

Exercise 18.3

The idea. The product rule with one smooth factor.

Step 1. $X = t$ ($dX = dt$, no dW), $Y = W$: $d(tW) = t\,dW + W\,dt + dt\,dW = t\,dW + W\,dt$.

Step 2. Integrate: $tW_t = \int_0^t s\,dW_s + \int_0^t W_s\,ds$, so $tW_t - \int_0^t W_s\,ds = \int_0^t s\,dW_s$, an Itô integral: a martingale (equivalently, $d(tW - \int W) = t\,dW$, with no dt term).

Step 3. Its mean is 0, so $E[1 \cdot W_1] = E\int_0^1 W_s\,ds$, and $E[W_1] = 0$. Hence $E\int_0^1 W_s\,ds = 0$.

Exercise 18.4

The idea. Collect dt terms; they cancel. Then take expectations.

$d(W^4) = 4W^3\,dW + 6W^2\,dt$; $d(tW^2) = W^2\,dt + t(2W\,dW + dt)$; $d(3t^2) = 6t\,dt$. So

$$\begin{aligned} d(W^4 - 6tW^2 + 3t^2) &= 4W^3\,dW + 6W^2\,dt - 6W^2\,dt - 12tW\,dW - 6t\,dt + 6t\,dt \\ &= (4W^3 - 12tW)\,dW. \end{aligned}$$

No dt term: a martingale, starting at 0. So $E[W_t^4] - 6tE[W_t^2] + 3t^2 = 0$, giving $E[W_t^4] = 6t^2 - 3t^2 = 3t^2$.

Exercise 18.5

The idea. Each factor is an exponential martingale; the product rule's cross term depends on the correlation.

Step 1. $dX = X\,dW$ and $dY = Y\,dB$ (Lesson 18: no dt terms).

Step 2. $d(XY) = X\,dY + Y\,dX + dX\,dY = XY\,dB + XY\,dW + XY\,dW\,dB$.

Independent: $dW\,dB = 0$, so $d(XY) = XY(dW + dB)$: no dt term, a martingale.

Correlation ρ: $dW\,dB = \rho\,dt$, so $d(XY) = \rho XY\,dt + XY(dW + dB)$: a drift ρXY. Then XY is not a martingale (for $\rho \neq 0$), but $e^{-\rho t}XY$ is. Correlated exponentials reinforce each other on average.

Exercise 18.6

The idea. The product rule with the discount factor (smooth, no dW).

$d(e^{-rt}S) = e^{-rt}\,dS - re^{-rt}S\,dt = (\mu - r)e^{-rt}S\,dt + \sigma e^{-rt}S\,dW$. The dt term vanishes only if $\mu = r$.

With $\mu = 8\%$: $E[e^{-r}S_1] = e^{-r}S_0e^{\mu} = 100e^{0.05} = 105.1271$: above 100, a submartingale. The share is expected to beat cash by 5% a year.

Exercise 18.7

The martingale test by groups, for W^3 and $W^3 - 3tW$.

```
import numpy as np

rng = np.random.default_rng(187)
n = 400_000
W_h = rng.normal(0, math.sqrt(0.5), n)                    # W_0.5
W_1 = W_h + rng.normal(0, math.sqrt(0.5), n)              # W_1 = W_0.5 + an independent step
edges = np.quantile(W_h, np.linspace(0, 1, 6))
group = np.clip(np.digitize(W_h, edges[1:-1]), 0, 4)      # five groups by W_0.5
print("group   mean W_0.5   gap for W^3   gap for W^3 - 3tW")
for k in range(5):
    s = group == k
    gap_a = (W_1[s] ** 3).mean() - (W_h[s] ** 3).mean()
    gap_b = (W_1[s] ** 3 - 3 * W_1[s]).mean() - (W_h[s] ** 3 - 1.5 * W_h[s]).mean()
    print(f"{k + 1:5d}   {W_h[s].mean():10.3f}   {gap_a:11.3f}   {gap_b:17.3f}")
```

What it prints.

```
group   mean W_0.5   gap for W^3   gap for W^3 - 3tW
    1       -0.990        -1.469               0.012
    2       -0.374        -0.567              -0.007
    3        0.002         0.007               0.009
    4        0.377         0.546               0.001
    5        0.991         1.504               0.004
```

Reading it. For W^3 the gap in each group is about $1.5\times$ (the group's $W_{0.5}$): negative for paths that are low at time 0.5, positive for high ones. That is the integrated drift, $E[\int_{0.5}^{1} 3W_t\,dt \mid W_{0.5}] = 3W_{0.5} \times 0.5$. The gaps average out over all groups, which is why $E[W^3]$ stays at 0 while W^3 is not a martingale. For $W^3 - 3tW$ every gap is noise.

Exercise 18.8

Reweighting by Z shifts W by θT.

```
theta, T = 0.5, 1.0
WT = rng.normal(0, math.sqrt(T), 1_000_000)
Z = np.exp(theta * WT - 0.5 * theta**2 * T)
for a in (0.0, 0.5, 1.0):
    lhs = (Z * (WT > a)).mean()
    rhs = 1 - Ncdf((a - theta * T) / math.sqrt(T))       # P(W_T + theta T > a)
    print(f"a = {a}: E[Z 1(W_T > a)] = {lhs:.4f},  P(W_T + theta T > a) = {rhs:.4f},  P(W_T >
a) = {1 - Ncdf(a):.4f}")
```

What it prints.

```
a = 0.0: E[Z 1(W_T > a)] = 0.6911,  P(W_T + theta T > a) = 0.6915,  P(W_T > a) = 0.5000
a = 0.5: E[Z 1(W_T > a)] = 0.4994,  P(W_T + theta T > a) = 0.5000,  P(W_T > a) = 0.3085
a = 1.0: E[Z 1(W_T > a)] = 0.3083,  P(W_T + theta T > a) = 0.3085,  P(W_T > a) = 0.1587
```

Reading it. In each case the weighted probability matches the probability for the shifted variable $W_T + \theta T$, not the original one. This is Girsanov's theorem at one time point; Lesson 25 extends it to the whole path.

Self-test

1. $c = \frac{1}{2} \times 0.3^2 = 0.045$.

2. Its dt term is $3W\,dt$: given where the path is, its future average is not its current value (higher when $W > 0$, lower when $W < 0$). The effects cancel overall, so its unconditional mean stays at 0, but a martingale needs the conditional statement.

3. $d(XY) = X\,dY + Y\,dX$: the cross term $dX\,dY$ is 0 when X has no dW part.